

CrewRoll encryption brief

Version 1.0.4, iOS build 20 | Prepared for Apple classification review

This document records implementation facts. It is not a BIS-issued CCATS, a French declaration, an ECCN assignment or an export-exemption certification.

App Store app	CrewRoll: Shared Trip Photos / 6797897853
Bundle / build	app.crewroll.mobile / 1.0.4 (20)
Apple build ID	c32d49ce-79e0-4d03-94b2-b26f9a139041
Source revision	a800a77a448f506a8fbc4425a2b2b058d509696f
Support case	102974597698 / opened 24 September 2026

Product and data flow

CrewRoll is a consumer app for private trip photo sharing between approved iPhone and Android members. Users take pictures with their phone's normal camera. Eligible photos are encrypted on the sending device, staged as encrypted data by the relay, decrypted on an approved recipient's device and saved to its photo library. The service separately processes readable account and trip metadata.

Application-layer cryptography

Photos and previews	libsodium crypto_secretstream_xchacha20poly1305. 256-bit keys; 24-byte stream header; 256 KiB plaintext chunks; 17-byte authentication overhead per chunk. The reader requires a valid final tag.
Photo manifest	crypto_aead_xchacha20poly1305_ietf. 256-bit key, fresh 192-bit nonce and 16-byte authentication tag. The encrypted manifest contains the per-photo content root and descriptors.
Trip-key envelopes	crypto_box_seal / crypto_box_seal_open. The bundled crypto_box primitive is Curve25519/XSalsa20-Poly1305; 32-byte public/private keys and 48-byte sealed-box overhead.
Key derivation	crypto_kdf_derive_from_key (BLAKE2b). Fresh random 32-byte content roots; separate CRROLL01 subkeys for preview and original; a CRMANF01 subkey from the trip key protects manifests.
Integrity	SHA-256 hashes cover plaintext and ciphertext. The receiver verifies framing, authentication, final tag, length and hashes before accepting a received original.

Implementation and review scope

Technical facts and questions for case 102974597698

Library and platform boundaries

The iOS source bundles libsodium 1.0.22, identified by the vendored header. CrewRoll calls library primitives rather than reproducing their internals. Its own versioned format defines message framing, authenticated context and parsing rules; this brief does not claim that the complete CrewRoll protocol is publicly standardized. Apple platform services provide HTTPS/TLS, Keychain storage and Secure Enclave P-256 identity keys.

Why clarification is requested

Libsodium documents XChaCha20-Poly1305 as widely implemented but not standardized [1]. Apple's questionnaire refers to algorithms not accepted by international standards bodies, while its export-compliance overview also discusses whether functionality has otherwise been published [2]. Lack of formal standardization alone does not resolve the complete application's export classification.

Questions for Apple

1. Which App Store Connect algorithm category applies to this published-library implementation, including its app-specific format?
2. Does Apple require a BIS-issued CCATS for this app, or another declaration or supporting document? Please identify the exact document and submission route.
3. The account holder approved excluding France from the initial Apple release. Please clarify which documents apply with France excluded and what would be needed before adding it later.

Current release state and next step

As checked on 26 September 2026, Apple has processed build 20 as VALID, with MISSING_EXPORT_COMPLIANCE for internal and external testing. No new classification or government document has been submitted. Apple's documentation distinguishes its encryption review from build processing [3]. If formal BIS classification is necessary, its published process uses SNAP-R and product technical documentation [4]; this brief can support that process but cannot replace its official result.

Source traceability

Verified against the release revision above: packages/contracts/crypto/FORMAT_V1.md; NativePhotoCrypto.swift; AppleSodiumCrypto.swift; AppleNativeKeyInfrastructure.swift; and the vendored Clibsodium version.h, crypto_box.h and crypto_kdf.h headers. Relevant implementation files are unchanged between the release revision and the repository inspected on 26 September 2026.

References

1. [Libsodium: authenticated encryption algorithms](#)
2. [Apple: overview of export compliance](#)
3. [Apple: required encryption documentation](#)
4. [BIS: encryption review and CCATS process](#)